MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR,
BI-FET OPERATIONAL AMPLIFIERS,
MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/114A, dated 9 November 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.1, line 2: Delete "Three" and substitute "Two".

PAGE 3

* 3.6, delete and substitute the following:

"3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

Add new paragraph 4.2.1 as follows:

"4.2.1 Qualification extension. For qualification inspection, if a manufacturer qualifies to device types 04, 05 and 06 which are designed and manufactured identically (same die, same process, same screening) in all respects (except electrical testing) to device types 01, 02 and 03 respectively, then qualification may be extended to device types 01, 02 and 03 when authorized by the qualifying activity. Additionally, Part I qualification may be extended to device types 01, 02 and 03 only after acceptance by the qualifying activity of subgroup C1 testing performed on these device types and submission of data in accordance with MIL-M-38510, appendix D."

PAGE 6

* TABLE I, settling time test, max limits column, devices 01 and 04 only: Delete "1500" and substitute "4000".

NOTES: 1/, line 3: Delete sentence "The measurements for bias currents must be made within 25 ms after power is first applied to the device for test.", and substitute "The measurements for bias currents must be made within 25 ms or 5 loop time constants, whichever is greater, after power is first applied to the device for test."

PAGE 8

* TABLE II: delete "Class C devices" column in its entirety.

* 4.4.1c., delete and substitute: "c. Subgroup 12 shall be added to group A inspection for class S devices only as shown in table III herein. The LTPD for subgroup 12 shall be 5."

PAGE 9

FIGURE 1: Add terminal connections for case H as printed on page 3 of this amendment.

PAGE 12

FIGURE 2, circuit C, delete and substitute new circuit C as printed on page 4 of this amendment

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PAGE 13

FIGURE 2: Add new circuit E as printed on page 5 of this amendment.

PAGE 20

FIGURE 9, Note 5, second line, delete and substitute the following:

"node is not nulled relative to the specified percentage of the final output value (See table I settling time test conditions)".

PAGE 25

* TABLE III, subgroup 12, tests 92 and 93, devices 01 and 04 only, max limits column: Delete "1500" and substitute "4000".

PAGE 27

* 4.4.3a, add the following sentence:

"Delta limits shall apply only to subgroup 1 of group C inspection for class B devices."

4.4.3 b and c; delete and substitute the following:

"b. Life test for class B devices (method 1005 of MIL-STD-883). Test condition D using the circuit shown on figure 4 or test condition C using the circuit shown on figure 5 or test condition F using the circuit shown on figure 6 (see note for 4.3a, item 2).

c. Subgroups 3 and 4 shall be added to group C inspection and shall consist of group A, subgroups 8 and 12, as specified in table III herein. The LTPD for subgroup 3 shall be 10 for class B; the LTPD for subgroup 4 shall be 5 for class B."

* 4.6, delete in its entirety.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - MI, AR
Air Force - 99
NASA - NS

User activities:
Army - SM
Navy - CG, MC, AS, OS, SH

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0615)

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Device types 01, 02, 03, 04, 05, and 06

(Case H)

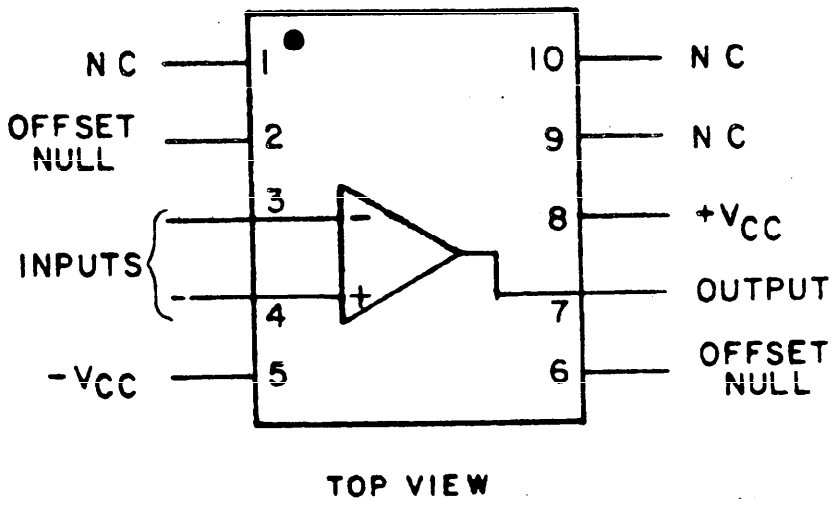
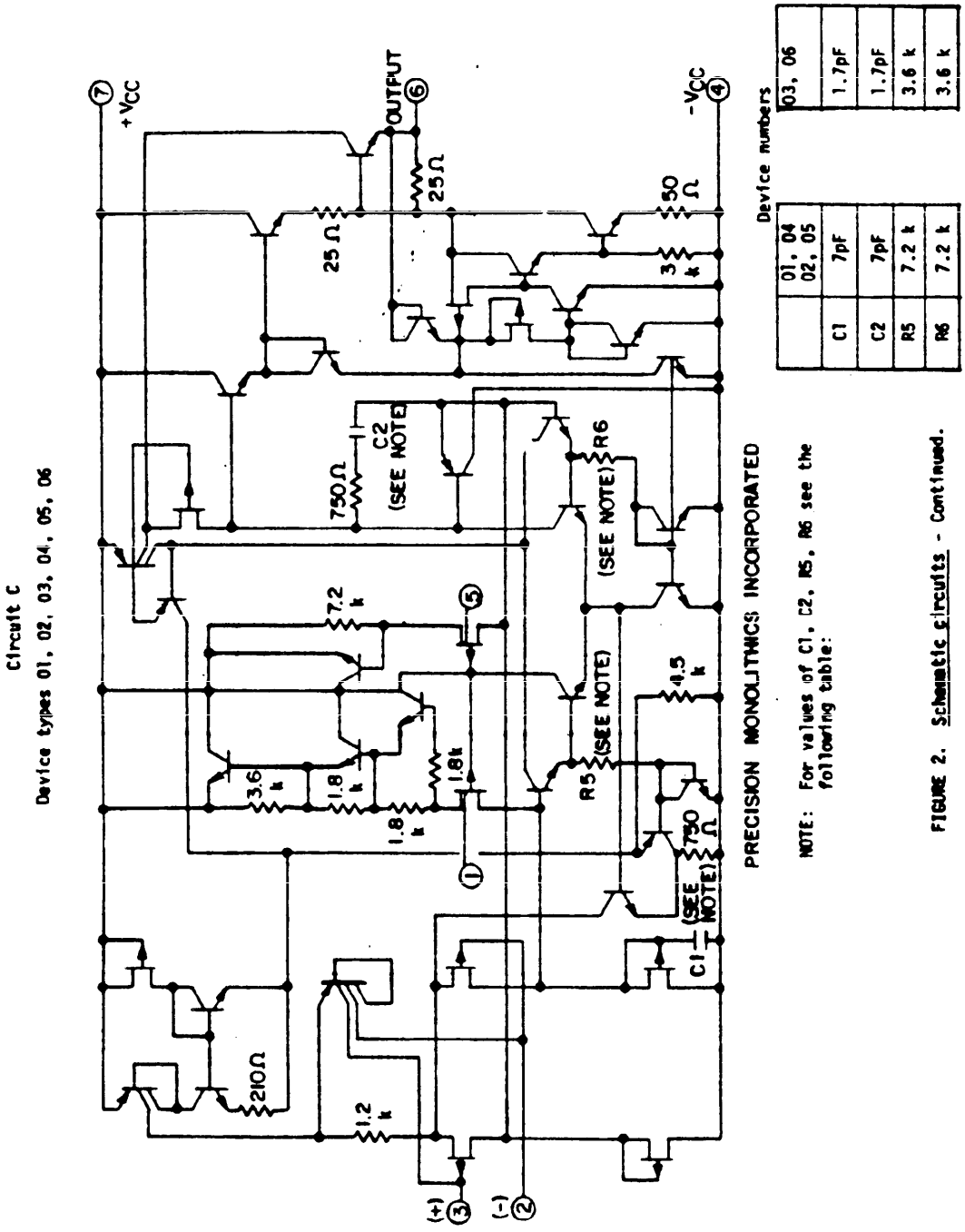


FIGURE 1. Terminal connections - Continued.

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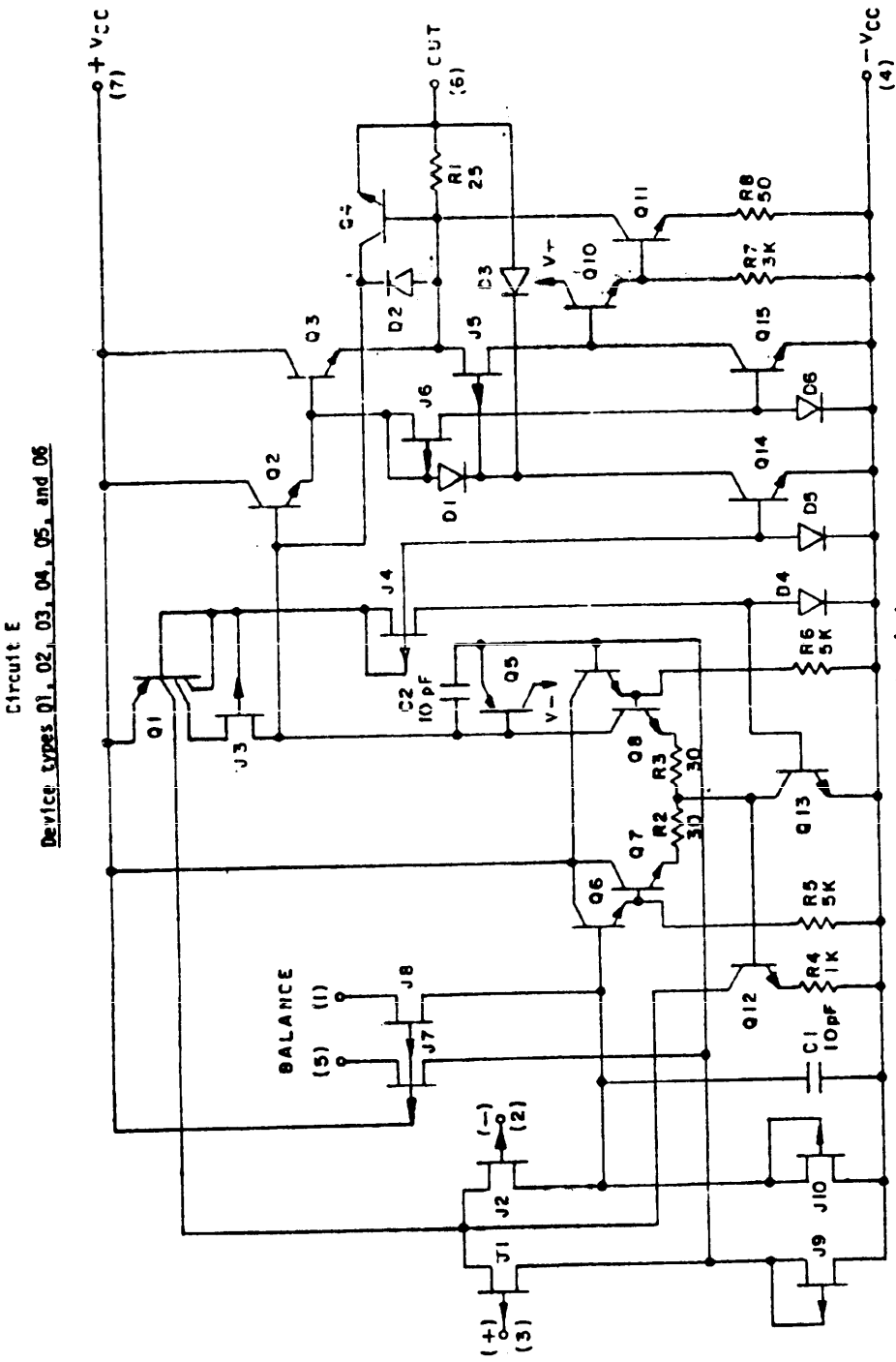


Device numbers	01, 04 02, 05	03, 06
C1	7pF	1.7pF
C2	7pF	1.7pF
R5	7.2 k	3.6 k
R6	7.2 k	3.6 k

NOTE: For values of C1, C2, R5, R6 see the following table:

FIGURE 2. Schematic circuits - Continued.

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NOTES:
1. For device types 03 and 06 C2 = 2 pF.
2. Component values are nominal. Resistance values are in ohms.

FIGURE 2. Schematic circuits - Continued.

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